The role of verbal working memory in predictability effects during reading

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Background

- **RQ**: Do predictability effects during reading reflect a resource-intensive predictive mechanism or resource-free pre-activation?
- Both mechanisms likely contribute to the predictability effects (Huettig, 2015; Pickering & Gambi, 2018)

Resource-free pre-activation mechanism

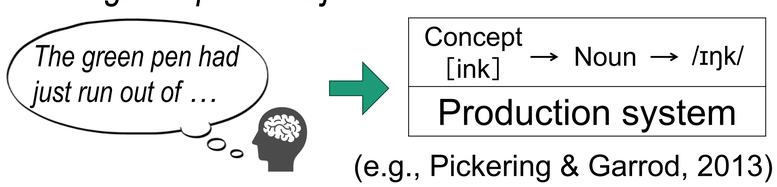
The green pen had just run out of …

red → colour; leaf; write; paper; ink

(e.g., Kukona et al. 2011)

Resource-intensive production-based mechanism

The green pen had just run out of …



(e.g., Pickering & Garrod, 2013)

- The use of the resource-intensive mechanism may be optional and hinge on readers' working memory (WM)

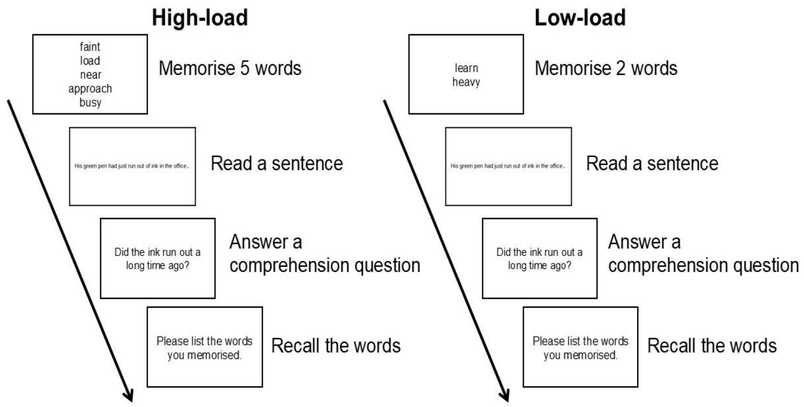
Experiment predictions

- If a resource-intensive mechanism drives the predictability effects, these effects should be reduced under high (vs. low) WM load
- This interaction may be more pronounced in lower-WM readers, as they should be more susceptible to WM load than higher-WM readers

Methods: Eye-tracking reading

- 2 (Predictability) × 2 (Load) within-subjects design
- 81 native English-speaking participants, 88 items

Predictability	Sentence
High-cloze (*M* cloze = .95)	His green pen had just run out of **ink** in the office.
Low-cloze (*M* cloze = 0)	The supply cabinet had just run out of **ink** in the office.

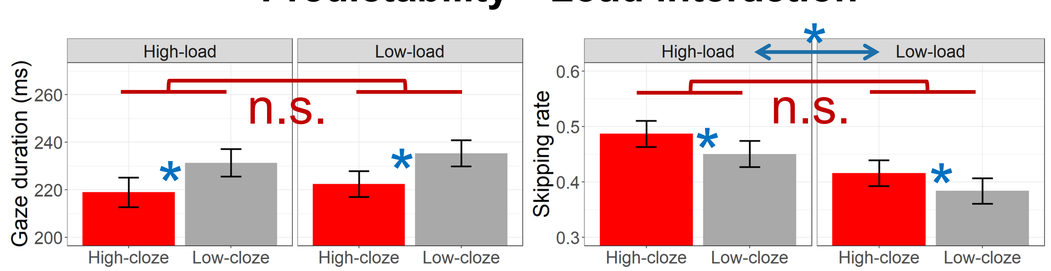


Results

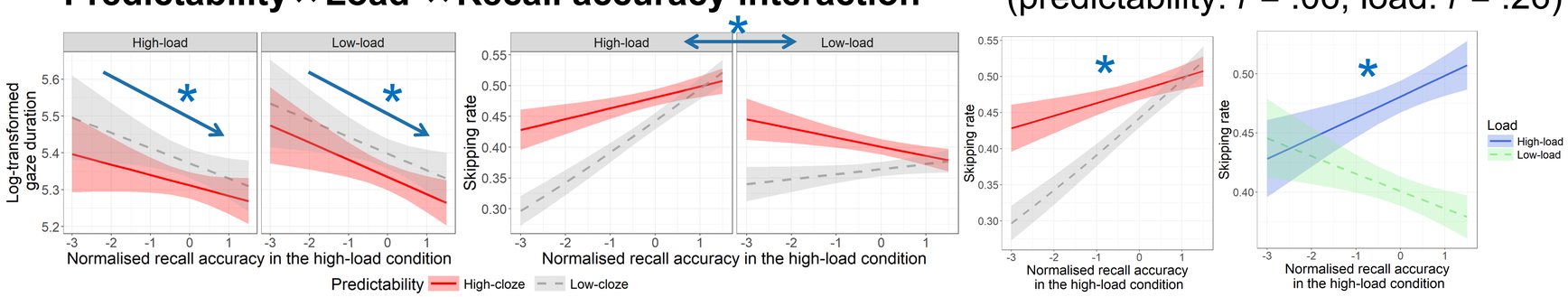
Recall accuracy: High-load 78% (*SD* = 13), Low-load 93% (*SD* = 6)

Comprehension accuracy: High-load 90% (*SD* = 29), Low-load 93% (*SD* = 25)

Predictability × Load interaction



Predictability × Load × Recall accuracy interaction



Overall:

- No Predictability × Load interaction on gaze duration or skipping rate

Recall effects on skipping rate:

- Predictability × Recall interaction
- Load × Recall interaction
- But the split-half reliability was low (predictability: *r* = .06, load: *r* = .26)

Conclusions

- The WM load did not modulate the predictability effects ➔ The predictability facilitation during reading is largely driven by a resource-free mechanism
- Shorter gaze durations and greater sensitivity to load (skipping) in higher-WM readers (vs. lower-WM readers) ➔ Strategic allocation of resources between the recall and reading tasks

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